Amit Kanubhai Patel

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Born in Chicago, IL, USA
February 27, 2026

Education

- **Doctor of Philosophy in Computer Science**, *Duke University*, May 2010. Advisor: Herbert Edelsbrunner.
- **Visiting Graduate Student**, *Institute of Science and Technology Austria*, Klosterneuburg, Austria, August 2009 – January 2010.
- **Visiting Graduate Student**, *Berlin Mathematical School*, Berlin, Germany, August 2007 – July 2008.
- **Master of Science in Computer Science**, *University of Illinois at Urbana–Champaign*, May 2003. Advisor: Jeff Erickson.
- **Bachelor of Science in Computer Science**, *University of Illinois at Urbana–Champaign*, May 2001.

Academic Appointments

Global footprint: USA, UK, France, Austria, Germany.

- **Associate Professor of Mathematics** (tenured), *Colorado State University*, Fort Collins, CO, USA, July 2022 – present.
- **Leverhulme Visiting Professor**, *School of Mathematical Sciences, Queen Mary University of London*, London, UK, September 2022 – August 2023.
- **Assistant Professor of Mathematics**, *Colorado State University*, Fort Collins, CO, USA, July 2016 – June 2022.
- **Member**, *School of Mathematics, Institute for Advanced Study*, Princeton, NJ, USA, September 2014 – June 2016.
- **Postdoctoral Fellow**, *Institute for Mathematics and its Applications, University of Minnesota*, Minneapolis, MN, USA, September 2013 – August 2014.
- **Postdoc and Coadjunct Professor**, *Department of Mathematics, Rutgers University*, New Brunswick, NJ, USA, September 2011 – June 2013.
- **Postdoctoral Researcher**, *GEOMETRICA, INRIA–Saclay*, Paris–Saclay, France, September 2010 – August 2011.
- **Postdoctoral Researcher**, *Institute of Science and Technology Austria*, Klosterneuburg, Austria, May 2010 – August 2010.

- **Intern**, *Lawrence Livermore National Laboratory*, Livermore, CA, USA, May 2006 – August 2006.

Industry & Engineering Experience (pre-PhD)

- **Software Engineer (DSP/Embedded Systems)**, *Texas Instruments*, Houston, TX, USA, May 2000 – December 2001.
  - Evaluated TCP/IP stacks for embedded DSP; recommended lean feature sets to reduce code size.
  - Implemented CSL peripheral drivers (DMA, McBSP, Timer, GPIO); built a chip-pin GUI shipped in Code Composer Studio 2.0.
  - Created a C code generator for 54x/55x device configurations; deployed HTTP/FTP/DNS/DHCP/Telnet on embedded devices.
- **Software Engineer**, *U.S. Army Corps of Engineers*, Champaign, IL, USA, June 1999 – May 2000.

Honors and Fellowships

- **Leverhulme Visiting Professorship**, *The Leverhulme Trust*, London, UK — 2022–2023.
- **Postdoctoral Fellowship**, *Institute for Mathematics and its Applications*, Minneapolis, MN, USA — 2013–2015.

Publications

1. Amit Patel and Primoz Skraba. *Möbius Homology.* arXiv, 2023. To appear in *Transactions of the American Mathematical Society*, 2026.
2. Amit Patel and Tatum Rask. *Poincaré Duality for Generalized Persistence Diagrams of (co)Filtrations. Journal of Applied and Computational Topology*, Volume 6, 2024.
3. Brittany Terese Fasy and Amit Patel. *Combinatorial Persistent Homology Transform. Foundations of Data Science*, Volume 6, Issue 3, 2024.
4. Alexander McCleary and Amit Patel. *Edit Distance and Persistence Diagrams over Lattices. SIAM Journal on Applied Algebra and Geometry*, Volume 6, Issue 2, 2022.
5. Robert MacPherson and Amit Patel. *Persistent Local Systems. Advances in Mathematics*, Volume 386, August 2021.
6. Vidit Nanda and Amit Patel. *Canonical Stratifications along Bisheaves. Proceedings of the Abel Symposium 2018: Topological Data Analysis*, 2020.
7. Alex McCleary and Amit Patel. *Bottleneck Stability for Generalized Persistence Diagrams. Proceedings of the AMS*, Volume 148, Number 7, pp. 3149–3161, 2020.

8. Justin Curry and Amit Patel. *Classification of Constructible Cosheaves. Theory and Applications of Categories*, Volume 35, Number 27, pp. 1012–1047, 2020.

9. Amit Patel. *Generalized Persistence Diagrams. Journal of Applied and Computational Topology*, Volume 1, Issue 3–4, June 2018.

10. Vin de Silva, Elizabeth Munch, and Amit Patel. *Categorified Reeb Graphs. Discrete & Computational Geometry*, Volume 55, Issue 4, pp. 854–906, June 2016.

11. Paul Bendich, Herbert Edelsbrunner, Dmitriy Morozov, and Amit Patel. *Homology and Robustness of Level and Interlevel Sets. Homology Homotopy Appl.*, Volume 15, Number 1, pp. 51–72, 2013.

12. Frédéric Chazal, Amit Patel, and Primoz Skraba. *Computing Well Diagrams for Vector Fields on $\mathbb{R}^n$. Applied Mathematics Letters*, Volume 25, Issue 11, pp. 1725–1728, November 2012.

13. Herbert Edelsbrunner, Dmitriy Morozov, and Amit Patel. *Quantifying Transversality by Measuring the Robustness of Intersections. Foundations of Computational Mathematics*, Volume 11, Issue 3, June 2011.

14. Herbert Edelsbrunner, Dmitriy Morozov, and Amit Patel. *The Stability of the Apparent Contour of an Orientable 2-Manifold*. In *Topological Methods in Data Analysis and Visualization: Theory, Algorithms, and Applications*, eds. V. Pascucci, X. Tricoche, H. Hagen, and J. Tierny, Springer-Verlag, Heidelberg, Germany, 2011.

15. Amit Patel. *Reeb Spaces and the Robustness of Preimages.* PhD thesis, Duke University, May 2010.

16. Paul Bendich, Herbert Edelsbrunner, Micheal Kerber, and Amit Patel. *Persistent Homology Under Non-Uniform Error.* In *Proceedings of the 35th International Symposium on Mathematical Foundations of Computer Science*, pp. 12–23, 2010.

17. Paul Bendich, Herbert Edelsbrunner, Dmitriy Morozov, and Amit Patel. *Robustness of Level Sets*. In *Proceedings of the 18th Annual European Symposium on Algorithms*, pp. 1–10, 2010.

18. Herbert Edelsbrunner, John Harer, and Amit Patel. *Reeb Spaces of Piecewise Linear Mappings.* In *Proceedings of the 24th Annual Symposium on Computational Geometry*, pp. 242–250, 2008.

Preprints and Submitted

1. Hideto Asashiba, Amit Patel. *Minimal Projective Resolutions, Möbius Inversion, and Bottleneck Stability.* arXiv, 2026.

2. Aziz Gülen, Facundo Mémoli, and Amit Patel. *ℓ^p-Stability of Weighted Persistence Diagrams.* arXiv, 2025.

3. Alex Elchesen and Amit Patel. *A Categorical Approach to Möbius Inversion via Derived Functors.* arXiv, 2024. Under review.

4. Dmitriy Morozov and Amit Patel. *Output-sensitive Computation of Generalized Persistence Diagrams for 2-filtrations.* arXiv, 2021.

Invited Talks (40 talks across 10 countries, 2011–2025)

Global footprint: invited talks delivered in the USA, United Kingdom, Canada, Germany, Brazil, Austria, Spain, Poland, Norway, and Uruguay; plus multiple international webinars.

1. *Möbius Homology* (August 18–22, 2025). The Geometric Realization of AATRN (Applied Algebraic Topology Research Network); Chicago, IL, USA.
2. *Two Languages, One Invariant: Unifying Möbius Homology and Betti Tables* (July 21–25, 2025). ATMCS — Algebraic Topology: Method, Computation, and Science; Bozeman, MT, USA.
3. *Möbius Inversions and Persistence* (March 27–28, 2023). London Taught Course Centre (LTCC) Intensive Courses; Department of Mathematics; University College London; London, UK.
4. *Möbius Inversions and Persistence* (February 16, 2023). Applied Topology Center; Oxford University; Oxford, UK.
5. *Algorithms for Generalized Persistence Diagrams* (November 1, 2021). Computational Persistence Homology; Online.
6. *Functoriality of Generalized Persistence Diagrams* (October 6, 2021). Topology Seminar; Northeastern University; Boston, MA, USA.
7. *Edit Distance and Persistence Diagrams Over Lattices* (May 1–3, 2021). Topological Data Analysis – Theory and Applications; Online.
8. *Persistent Local Systems* (June 15–18, 2020). Workshop on Topological Data Analysis (moved online due to COVID); Fields Institute; Toronto, ON, Canada.
9. *Möbius Inversions and Persistent Homology* (December 6–10, 2019). Canadian Mathematical Society Winter Meeting; Toronto, ON, Canada.
10. *Persistent Homology of Level Sets* (August 5–9, 2019). Summer School on Persistent Homology and Barcodes; JLU Gießen — Schloss Rauischholzhausen; Rauischholzhausen, Germany.
11. *Multiparameter Persistent Homology: A New Algebraic Framework* (June 17–19, 2019). Workshop on Topological Data Analysis; UNESP Campus de Rio Claro; Rio Claro, Brazil.
12. *Tensors in Multiparameter Persistent Homology* (June 3–7, 2019). Tensors: Algebra, Computation, and Applications; University of Colorado Boulder; Boulder, CO, USA.
13. *Persistent Local Systems (main talk)* (February 26, 2019). Topology Seminar; University of Colorado Boulder; Boulder, CO, USA.
14. *Persistence Beyond Vector Spaces* (February 5–7, 2019). Expository Workshop "Persistent Homology on Metric Spaces"; Department of Mathematics; The Ohio State University; Columbus, OH, USA.

15. *My Time with Herbert Edelsbrunner* (June 20–21, 2018). IST Austria 10; Institute of Science and Technology Austria; Klosterneuburg, Austria.

16. *Persistent Local Systems* (June 4–8, 2018). Abel Symposium: Topological Data Analysis; Geiranger, Norway.

17. *Persistent Local Systems* (May 21–25, 2018). Bridging Statistics and Sheaves; Institute for Mathematics and its Applications; Minneapolis, MN, USA.

18. *Generalized Persistence Diagrams* (August 1, 2017). Topological Data Analysis: Developing Abstract Foundations; Banff International Research Station; Banff, AB, Canada.

19. *Generalized Persistence Diagrams* (July 10, 2017). Foundations of Computational Mathematics; Barcelona, Spain.

20. *Generalized Persistence Diagrams* (January 31, 2017). Applied Algebraic Topology Research Network (webinar); Online.

21. *Classification of Constructible Cosheaves* (January 4–5, 2017). AMS Special Session on Sheaves in Topological Data Analysis; Joint Mathematics Meetings; Atlanta, GA, USA.

22. *Semicontinuity of Persistence Diagrams* (October 2, 2016). SIAM Central States Section; Little Rock, AR, USA.

23. *Semicontinuity of Persistence Diagrams* (May 16, 2016). Topology, Geometry, and Data Analysis @ OSU; Columbus, OH, USA.

24. *Persistence for Maps* (June 15–20, 2015). Dynamics, Topology, and Computations; Bedlewo, Poland.

25. *Persistent Objects* (December 15–17, 2014). Foundations of Computational Mathematics Conference; Universidad de la República; Montevideo, Uruguay.

26. *The Persistent Homology Group* (December 4, 2014). Topology Seminar; Princeton University; Princeton, NJ, USA.

27. *Persistence for Maps to Manifolds* (September 15–19, 2014). Workshop on Generalized Persistence and Applications; American Institute of Mathematics; Palo Alto, CA, USA.

28. *The Quillen 2–Construction for Persistence* (May 26–30, 2014). Algebraic Topology — Methods, Computation and Science 6; Pacific Institute for the Mathematical Sciences; Vancouver, BC, Canada.

29. *Persistent Sheaves for Stratified Maps* (April 14, 2014). Topology Seminar; University of Minnesota; Minneapolis, MN, USA.

30. *Computing Well Diagrams for the Fixed Points of a Vector Field* (April 3, 2014). Geometry and Topology Seminar; Tulane University; New Orleans, LA, USA.

31. *Persistent Sheaves* (February 28, 2014). Topology, Geometry and Data Seminar; The Ohio State University; Columbus, OH, USA.

32. *Connecting Persistent Homology Groups* (October 7–11, 2013). Workshop on Topological Data Analysis; Institute for Mathematics and its Applications; Minneapolis, MN, USA.

33. *Measuring the Stability of Intersections to C^0 Perturbations* (August 8–11, 2013). SIAM Conference on Applied Algebraic Geometry; Fort Collins, CO, USA.

34. *Multidimensional Persistence and Sheaves* (April–May, 2013). MacPherson Seminar; Institute for Advanced Study; Princeton, NJ, USA.

35. *The Étalage of a Map* (October 14–19, 2012). Workshop on the Interface of Computational Topology and Machine Learning Theory; Banff International Research Station; Banff, AB, Canada.

36. *Sheaves and Persistence* (July 9–11, 2012). Applied Algebraic Topology: The Next Generation minisymposium; SIAM Annual Meeting (AN12), jointly with the SIAM Conference on Financial Mathematics and Engineering (FM12); Minneapolis, MN, USA.

37. *Well Groups* (June 17–20, 2012). Workshop on Computational Topology; Symposium on Computational Geometry (SoCG); Chapel Hill, NC, USA.

38. *Algebraic Well Groups* (October 6–9, 2011). SIAM Conference on Applied Algebraic Geometry; Raleigh, NC, USA.

39. *Reeb Spaces of Piecewise Linear Maps* (June 10, 2008). Symposium on Computational Geometry (SoCG); College Park, MD, USA.

Grants and Funding

- **NSF DMS: Topology**, *Developing Theory and Applications of Möbius Homology* — PI, $224,628 (Proposal submitted 9/15/2024; Denied 7/28/2025). Joint NSF–EPSRC project with Prof. Primoz Skraba, Queen Mary University London.

- **Leverhulme Trust**, *Leverhulme Visiting Professorship* — PI, £56,480 (9/2021–8/2022).

- **NSF CISE**, *REU Supplement* — PI, $6,000 (6/1/2020–8/1/2020).

- **NSF CISE**, *Reeb Spaces and Parameterized Clustering (sole PI)* — PI, $291,284 (8/15/2017–8/31/2021).

Teaching Experience

Year	Term	Course	Cr	Enroll
2025	Fall	M230 Intro to Mathematical Reasoning	3	28
	Fall	M571 Topology II	3	9
	Spring	M235 Intro to Mathematical Reasoning	2	29
	Spring	M570 Topology I	3	6
2024	Fall	M472 Introduction to Topology	3	11
	Spring	M571 Topology II	3	12
2023	Fall	M570 Topology I	3	15
	Fall	DSCI369 Linear Algebra for Data Science	4	120
2022	Spring	M571 Topology II	3	15
2021	Fall	M570 Topology I	3	10
	Fall	M235 Intro to Math Reasoning	2	36
	Spring	M261 Calc for Physical Scientists III	4	28
	Spring	M567 Intro to Abstract Algebra II	3	6
2020	Fall	M566 Intro to Abstract Algebra I	3	11
2019	Fall	M261 Calc for Physical Scientists III	4	90
	Fall	M261 Calc for Physical Scientists III	4	92
	Spring	M567 Intro to Abstract Algebra II	3	8
2018	Fall	M466 Abstract Algebra I	3	17
	Fall	M566 Intro to Abstract Algebra I	3	15
	Spring	M235 Intro to Math Reasoning	2	30
2017	Fall	M419 Intro to Complex Variables	3	23
	Spring	M567 Intro to Abstract Algebra II	3	11
2016	Fall	M566 Intro to Abstract Algebra I	3	9

Advisees

- Tatum Rask — Master of Science (Fall 2022); Ph.D. (expected Spring 2026).
- Alex Elchesen — Postdoc (2022–2025), Ph.D., University of Florida.
- Jacob Cleveland — Master of Science (Fall 2024).
- Alexander McCleary — Ph.D. (Summer 2021).
- Dustin Sauriol — Master of Science (Fall 2018).

Service and Outreach

- **Organizer**, *Bridging Algebraic Combinatorics, Topology and Applications (BACTA 2025)* — August 4–8, 2025, Rutgers University, NJ, USA.

- **Member**, Combinatorics/Topology Faculty Hiring Committee — 2024–2025. (To hire two tenure-track faculty in combinatorics or topology.)
- **Organizer**, Workshop on Möbius Inversions and Reeb Spaces — June 13, 2023, as part of Computational Geometry Week, Dallas, TX, USA.
- **Member**, Graduate Committee — Fall 2023–Fall 2025. (Oversees admissions, recruitment, and other administrative duties.)
- **Member**, Department Executive Committee — 2021–2022. (Advises department chair.)
- **Organizer**, CSU Category Theory Lab — 2018–2019. website
- **Organizer**, CSU Topology Seminar — 2017–present. website
- **Member**, Applied Category Theory — 2020–2022. website

Other Professional Activities

Editorial Roles

Editor for *Applied and Computational Topology and Geometry*, published by the Associates for Mathematical Research (AMR).